

Occult Blood, Fecal, Immunoassay (ColoFIT™)

Test Details

METHODOLOGY

Immunoassay (IA)

RESULT TURNAROUND TIME

1 - 3 days

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

Qualitative detection of fecal occult blood

Limitations

This test is intended only for the detection of human hemoglobin in fecal specimens. It is not for use in testing urine, gastric specimens, or other bodily fluids. Results cannot be considered conclusive evidence of the presence or absence of gastrointestinal bleeding or pathology. This test is designed for preliminary screening. A positive result should be followed up with additional diagnostic procedures, such as colonoscopy or sigmoidoscopy, to determine the exact cause and source of the occult blood in the feces. A negative result can be obtained even when a gastrointestinal disorder is present. For example, some polyps and colorectal cancers may bleed intermittently or not at all during certain stages of disease. False-negative results may occur when occult blood is not uniformly distributed throughout the stool. Repeat testing is recommended if a pathological condition is suspected. Urine and excessive dilution of specimens with water from the toilet bowl may cause erroneous results. For best results, use the collection paper in the collection kit.

Patients with menstrual bleeding, bleeding hemorrhoids, constipation bleeding, and urinary bleeding should not be considered for testing as these conditions may interfere with test results. These patients may be considered for testing after such bleeding ceases.

Alcohol and certain medications, such as aspirin, indomethacin, reserpine, phenylbutazone, corticosteroids, and nonsteroidal anti-inflammatory drugs, may cause gastrointestinal irritation and subsequent bleeding in some patients.

Custom Additional Information

A screening test for colorectal cancer, diverticulitis, colitis, polyps, and adenomas should be highly sensitive and highly specific for bleeding in the lower gastrointestinal tract, and should encourage patient compliance with obtaining a proper specimen. Guaiac-based tests for occult blood in stool suffer from a number of drawbacks, leading to a reduction in their sensitivity as well as production of false-positive results. These drawbacks include:

- Detection of heme present in the stool, which can originate from bleeding anywhere in the gastrointestinal tract
- Inability to distinguish human heme from heme present in many foods, such as dietary meats
- Indiscriminate detection of peroxidase activity, which is present in a wide range of fruits and vegetables
- Reaction with drugs and other substances that can cause false-negative results

Detection of fecal occult blood by immunoassay eliminates the drawbacks seen with guaiac-based tests, and provides a sensitivity and specificity that is superior to guaiac. Immunochemical fecal occult blood tests utilize antihuman hemoglobin antibodies to detect the globin portion of undegraded human hemoglobin in stool. While hemoglobin from the upper gastrointestinal tract is mostly degraded by bacterial and digestive enzymes before reaching the large intestine, hemoglobin from lower gastrointestinal tract bleeding undergoes less degradation and remains immunochemically reactive. Thus, these tests are specific for lower gastrointestinal tract bleeding. Additionally, immunochemical fecal occult blood tests do not react with animal hemoglobin; peroxidase compounds; or with drugs, vitamins, or other substances that can produce false results, thereby eliminating the need for dietary or drug restrictions prior to patient specimen collection.

The Polymedco OC-Auto Micro 80 immunochemical fecal occult blood test is capable of detection of as little as 100 ng of human hemoglobin per mL of feces, making it a much more sensitive assay than guaiac-based tests for detecting low levels of human fecal occult blood. It requires only one stool specimen collected from one bowel movement.

Specimen Requirements

Specimen

Stool in Polymedco OC-Auto Sampling Bottle

Volume

One Polymedco OC-Auto Sampling Bottle with stool

Container

Polymedco OC-Auto Sampling Bottle

Collection Instructions

Enter patient identification and date of collection on sampling bottle label. Place collection paper inside toilet bowl on top of water. Deposit fecal sample on top of collection paper. Do **not** allow fecal sample to contact toilet water until after specimen has been collected. Open the green cap on the sampling bottle by twisting and pulling upwards. Scrape the surface of the fecal sample with the sampling probe. Cover the grooved portion of the sampling probe completely with the fecal sample. Close the sampling bottle by inserting the sampling probe and snap the green cap on tight. Do not reopen. Flush remaining stool and used collection paper. Return the sampling bottle to the physician or laboratory. Only one stool specimen collected from one bowel movement is required.

Storage Instructions

Specimen in sampling bottle can be stored for 15 days at room temperature.

Patient Preparation

No special drug or dietary restrictions are required. Results of this test are not affected by dietary peroxidases, animal blood, or vitamin C.

Causes for Rejection

Collection tube or sampling device other than the Polymedco OC-Auto Sampling Bottle; specimens with obvious blood (which may be due to menstrual bleeding, bleeding hemorrhoids, constipation bleeding, or urinary bleeding); raw stool submitted in a container; unlabeled specimen or name discrepancy between specimen and test request label; specimen received after prolonged delay (more than 15 days at room temperature since collection); specimen received frozen; expired sampling bottle; leaking sampling bottle; sampling bottle overfilled or without adequate liquid

References

Fraser CG, Matthews CM, Mowat NA, Wilson JA, Carey FA, Steele RJ. Immunochemical testing of individuals positive for guaiac faecal occult blood test in a screening programme for colorectal cancer: An observational study. *Lancet Oncol.* 2006 Feb; 7(2):127-131 PubMed 16455476 (<http://www.ncbi.nlm.nih.gov/pubmed/16455476>)

Levin B, Brooks D, Smith RA, Stone A. Emerging technologies in screening for colorectal cancer: CT colonography, immunochemical fecal occult blood tests, and stool screening using molecular markers. *CA Cancer J Clin.* 2003 Jan-Feb; 53(1):44-55. PubMed 12568443 (<http://www.ncbi.nlm.nih.gov/pubmed/12568443>)

Skaife P, Seow-Choen F, Eu KW, Tang CL. A novel indicator for surveillance colonoscopy following colorectal cancer resection. *Colorectal Dis.* 2003 Jan; 5(1):45-48. PubMed 12780926 (<http://www.ncbi.nlm.nih.gov/pubmed/12780926>)